[5] Database and API Documentation

Project: Eventura (powered by Clientura) — Enterprise Event & Hackathon Platform
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Audience: Backend Engineers, Database Administrators, API Consumers, Integration Partners

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1. Database Architecture & Engine Overview

Eventura utilizes **PostgreSQL 15+** as its primary persistent relational datastore, managed and queried through **Prisma ORM**. The data tier enforces ACID transactions, foreign key constraints, composite unique indexes, and audit timestamps across all tables.

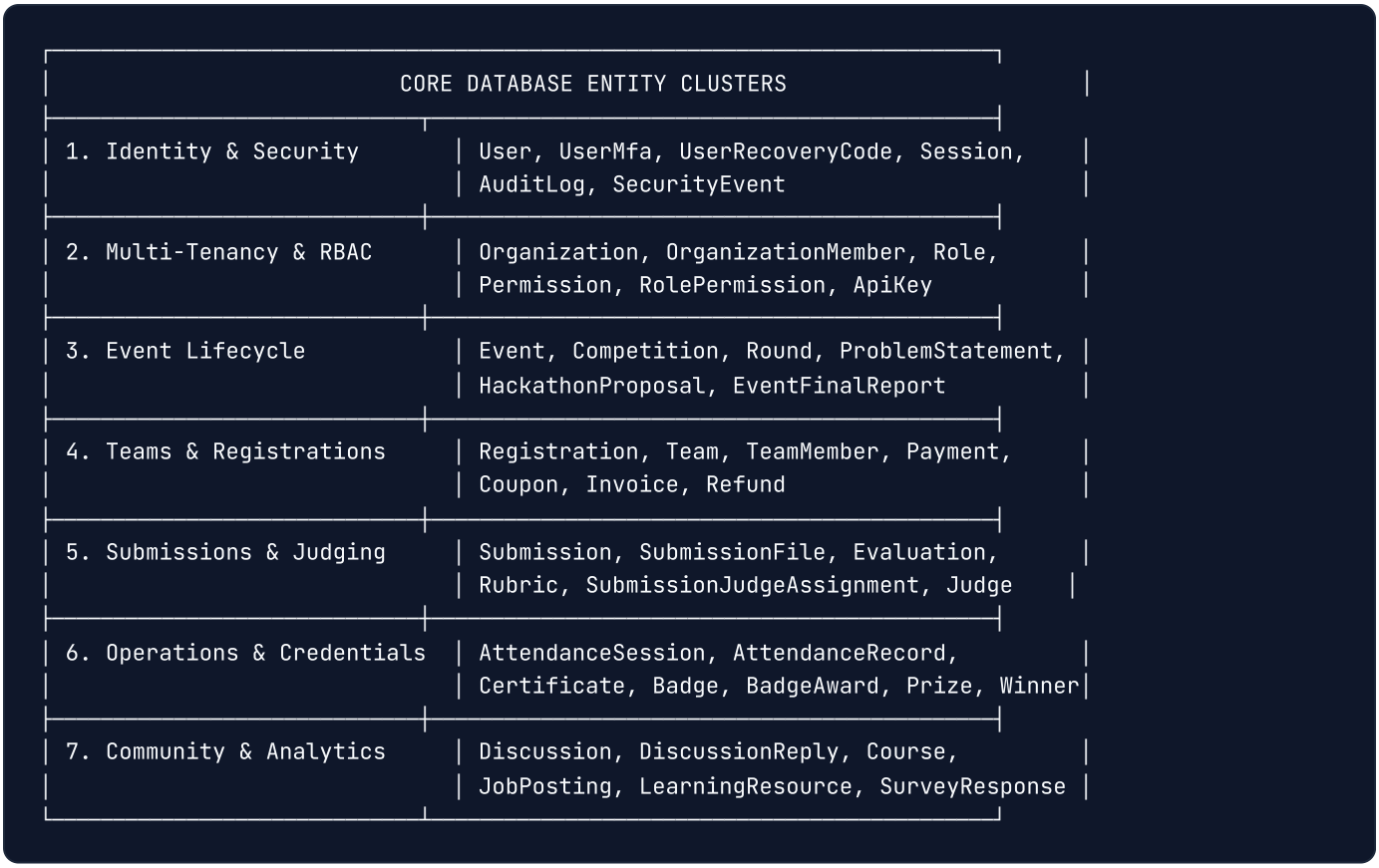
1.1 Key Database Characteristics

- **Relational Integrity**: Strict cascading rules (`ON DELETE CASCADE` or `RESTRICT`) to ensure orphan records cannot exist.
- **Multi-Tenant Composite Keys**: Critical tables enforce composite unique constraints (e.g. `[organizationId, slug]` , `[eventId, userId]`) to prevent duplicate records within the same tenant.
- **Audit Tracking**: Every domain entity contains `createdAt` and `updatedAt` timestamps, with soft-deletion support (`deletedAt`) on sensitive operational tables.
- **Migration Discipline**: All schema changes are tracked in forward-only SQL migration files in `prisma/migrations/` .

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2. Relational Entity Catalog

The database schema encompasses 45+ relational entities categorized into 7 core functional clusters:



2.1 Identity & Multi-Tenancy Schema Details

- **`User`**: Core identity table storing UUID, unique email, bcrypt `passwordHash`, first/last names, and operational status (`ACTIVE`, `PENDING`, `SUSPENDED`).
- **`UserMfa`**: Stores base32 TOTP secret, enrollment status, and last verification timestamp.
- **`Organization`**: Multi-tenant workspace entity storing UUID, human-readable name, unique URL slug, and lifecycle status.
- **`OrganizationMember`**: Join record mapping `userId` to `organizationId` with assigned `roleId`.
- **`Role`** & **`Permission`**: Decoupled RBAC definition allowing tenants to define custom access rules.

2.2 Event & Hackathon Competition Details

- **`Event`**: Stores event title, description, banner URL, venue/mode (IN_PERSON, VIRTUAL, HYBRID), dates, registration deadlines, max team sizes, and status (`DRAFT`, `PUBLISHED`, `REGISTRATION_OPEN`, `ONGOING`, `COMPLETED`).
- **`Competition`** & **`Round`**: Defines event tracks, multi-round elimination schedules, and specific judging rubrics.
- **`ProblemStatement`**: Technical briefs published per event. When a team registers their choice, `Team.problemStatementId` is locked.
- **`Submission`** & **`SubmissionFile`**: Stores project deliverables, GitHub repository links, demo URLs, file sizes, MIME types, and local/S3 storage paths.
- **`Evaluation`**: Quantitative scores and qualitative feedback recorded by assigned judges.

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3. RESTful API Specification

The Eventura backend exposes a uniform REST API mounted at `/api/v1`.

3.1 Global Conventions & Headers

- **Base URL:** `https://<API_DOMAIN>/api/v1`
- **Authentication:** `Authorization: Bearer <JWT_ACCESS_TOKEN>`
- **Tenant Scope:** `x-organization-id: <ORGANIZATION_UUID>` (Required for all tenant-scoped endpoints)
- **Content Type:** `Content-Type: application/json`

3.2 Standard Response Schemas

Success Response (HTTP 200 / 201):

```
{
  "success": true,
  "data": { ... },
  "message": "Operation completed successfully"
}
```

Error Response (HTTP 400 / 401 / 403 / 404 / 500):

```
{
  "success": false,
  "error": {
    "code": "MISSING_TENANT",
    "message": "x-organization-id header is required for this route",
    "details": []
  }
}
```

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4. Complete API Endpoints Catalog

Below is the definitive catalog of all 40 API route groups mounted in `server/src/routes/`:

4.1 Authentication & Profile (`/auth` , `/users`)

Method	Path	Auth	Roles / Permissions	Description
`POST`	`/auth/signup`	Public	None	Register new participant account
`POST`	`/auth/login`	Public	None	Authenticate with email/password; returns JWT or MFA challenge
`POST`	`/auth/mfa/verify`	Public	Temp MFA Token	Validate 6-digit TOTP code and issue session JWT
`POST`	`/auth/refresh`	Optional	Refresh Token	Issue new access token using refresh token
`POST`	`/auth/logout`	Required	Any Authenticated	Invalidate active user session
`GET`	`/users/me`	Required	Any Authenticated	Retrieve current user profile and memberships
`PUT`	`/users/me`	Required	Any Authenticated	Update current user profile details
`GET`	`/users`	Required	`users.read`	List organization users with pagination
`PUT`	`/users/:id/role`	Required	`users.manage`	Update a user's role in the organization

4.2 Multi-Tenant Organizations (`/organizations`)

Method	Path	Auth	Roles / Permissions	Description
`GET`	`/organizations`	Required	Any Authenticated	List organizations where user has active membership
`POST`	`/organizations`	Required	`org.create` / Sudo	Create a new organization tenant workspace
`GET`	`/organizations/:id`	Required	Member	Get details and settings for specific organization
`PUT`	`/organizations/:id`	Required	`org.manage`	Update organization settings and domain branding
`GET`	`/organizations/:id/members`	Required	`org.members.read`	List members and their assigned roles
`POST`	`/organizations/:id/members`	Required	`org.members.invite`	Invite new member to organization

4.3 Events & Competitions (`/events` , `/competitions` , `/hackathon-proposals`)

Method	Path	Auth	Roles / Permissions	Description
`GET`	`/events`	Tenant	Public / Member	List events scoped to tenant workspace
`POST`	`/events`	Required	`events.create`	Create a new event draft
`GET`	`/events/:id`	Tenant	Public / Member	Retrieve comprehensive event details, rounds, tracks
`PUT`	`/events/:id`	Required	`events.manage`	Update event information, schedule, or banners
`DELETE`	`/events/:id`	Required	`events.delete`	Soft delete event
`POST`	`/events/:id/publish`	Required	`events.publish`	Transition event status to `PUBLISHED`
`GET`	`/hackathon-proposals`	Required	`proposals.read`	List hackathon proposals submitted by coordinators
`POST`	`/hackathon-proposals`	Required	`proposals.create`	Submit new hackathon proposal with budget and tracks
`POST`	`/hackathon-proposals/:id/approve`	Required	`proposals.approve`	Approve proposal and spawn draft event
`POST`	`/hackathon-proposals/:id/reject`	Required	`proposals.reject`	Reject proposal with structured feedback
`GET`	`/competitions/:id/rounds`	Tenant	Member	List rounds and stage criteria for competition

4.4 Registrations, Teams & Payments (`/registrations` , `/teams` , `/payments`)

Method	Path	Auth	Roles / Permissions	Description
`POST`	`/registrations`	Required	Participant	Register for event (individual or team)
`GET`	`/registrations/me`	Required	Participant	Get current user's registration tickets and statuses
`GET`	`/registrations/event/:id`	Required	`events.manage`	List all registered attendees for an event
`POST`	`/teams`	Required	Participant	Create a new team and generate invite code
`POST`	`/teams/join`	Required	Participant	Join existing team using invite code
`POST`	`/payments/checkout`	Required	Participant	Create Stripe Checkout Session for paid event
`POST`	`/payments/webhook`	Public	Stripe Signature	Handle Stripe webhook events (`checkout.session.completed`)

4.5 Problem Statements & Submissions (`/problem-statements` , `/submissions`)

Method	Path	Auth	Roles / Permissions	Description
`GET`	`/problem-statements`	Tenant	Registered / Staff	List available problem statements for event
`POST`	`/problem-statements/select`	Required	Participant (Team Lead)	Permanently lock chosen problem statement for team
`POST`	`/submissions`	Required	Participant (Team Member)	Upload deliverable files, GitHub links, demo URL
`GET`	`/submissions/round/:roundId`	Required	Judge / Manager	List submissions submitted for specific round
`GET`	`/submissions/:id`	Required	Team / Judge / Staff	Get full submission details and attachments
`GET`	`/submissions/:id/files/:fileId/view`	Required	Team / Assigned Judge / Staff	**Stream submission file inline for safe view-only preview**

4.6 Judging, Evaluator Scoring & Leaderboard (`/evaluations` , `/judges`)

Method	Path	Auth	Roles / Permissions	Description
`GET`	`/judges/assigned`	Required	Judge	List competitions and submissions assigned to current judge
`POST`	`/judges/assign`	Required	`events.manage`	Assign judge to specific competition round
`POST`	`/evaluations`	Required	Judge	Submit rubric criteria scores and qualitative feedback
`GET`	`/evaluations/submission/:id`	Required	Judge / Manager	Get submitted scores for a submission
`GET`	`/evaluations/leaderboard/:eventId`	Tenant	Member	Get real-time computed leaderboard rankings

4.7 Attendance & Check-In (`/attendance`)

Method	Path	Auth	Roles / Permissions	Description
`POST`	`/attendance/sessions`	Required	`attendance.manage`	Create check-in session for event (e.g. Day 1 Morning)
`GET`	`/attendance/ticket/:regId`	Required	Participant	Get signed QR code token for event entry
`POST`	`/attendance/scan`	Required	Coordinator / Staff	Scan attendee QR code and record check-in timestamp
`GET`	`/attendance/session/:id/stats`	Required	`attendance.read`	Get real-time check-in counts and attendance rate

4.8 Certificates & Verification (`/certificates`)

Method	Path	Auth	Roles / Permissions	Description
`POST`	`/certificates/issue`	Required	`certificates.manage`	Issue digital certificates to event participants/winners
`GET`	`/certificates/me`	Required	Participant	List certificates awarded to current user
`GET`	`/certificates/:id/download`	Required	Recipient / Admin	Download generated vector PDF certificate
`GET`	`/certificates/verify/:code`	Public	None	**Public verification endpoint for certificate authenticity**

4.9 Reports & Analytics (`/reports` , `/analytics`)

Method	Path	Auth	Roles / Permissions	Description
`GET`	`/reports/dashboard`	Required	`reports.read`	Retrieve executive KPI metrics (revenue, attendees, teams)
`GET`	`/reports/export/:type`	Required	`reports.export`	Export CSV report for registrations, scores, or attendance
`GET`	`/analytics/events/:id`	Required	`analytics.read`	Real-time event engagement and submission metrics

4.10 AI Services (`/ai-copilot` , `/ai-validation`)

Method	Path	Auth	Roles / Permissions	Description
`POST`	`/ai-copilot/query`	Required	Participant / Mentor	Submit technical coding question to Gemini AI Copilot
`POST`	`/ai-validation/proposal`	Required	`proposals.manage`	AI risk and feasibility evaluation for hackathon proposal

4.11 Extended Platform Routes

The platform additionally mounts specialized routes for:

- `/badges` : Gamification badges, achievements, and criteria evaluations.
- `/winners` : Winner selection, podium distribution, and prize allocations.
- `/communications` & `/notifications` : Broadcast announcements and notification feeds.
- `/community` : Discussion forums, channels, and networking groups.
- `/learning` : Educational workshops, video modules, and resource libraries.

- `/recruitment`: Talent search and resume pool for event sponsors.
- `/sponsors`: Sponsor tier management and booth analytics.
- `/volunteers`: Volunteer scheduling and shift management.
- `/feedback`: Post-event feedback surveys and NPS scoring.
- `/developer` & `/security`: API key management, webhook logs, and security audit events.

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5. Schema Migration & Synchronization Strategy

1. **Deploying Schema Changes**:

```
npx prisma migrate deploy
```

2. **Generating Updated Types**:

```
npx prisma generate
```

3. **Idempotent RBAC Sync**: Whenever new route permissions are introduced, execute:

```
npx tsx prisma/scripts/sync-reports-permissions.ts
```
